Data Scientist Salaries Across Countries and Experience Levels

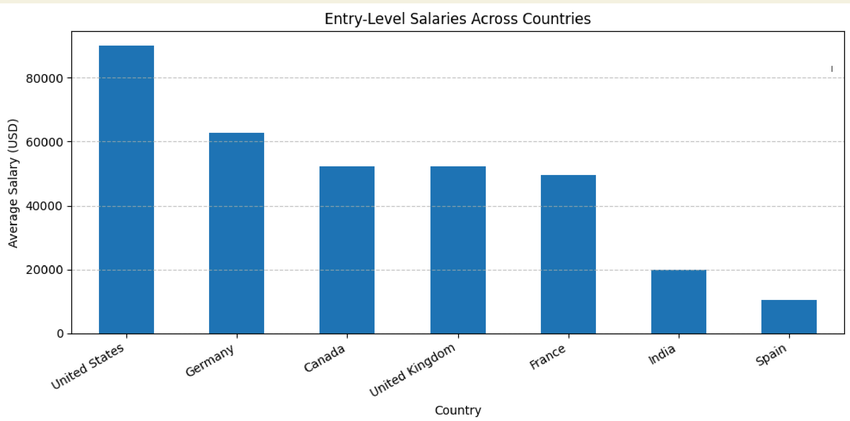
Abstract

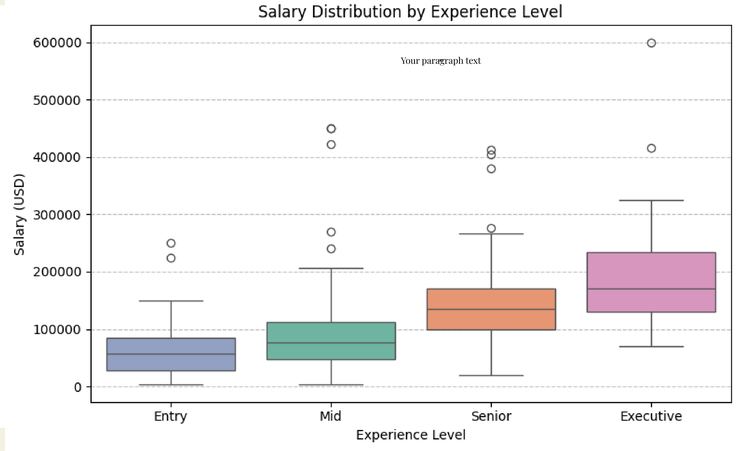
This project explores how data scientist salaries vary across countries and experience levels. Using a real-world dataset from Kaggle, I analyze salary trends and visualize geographic differences through an interactive map. The goal is to better understand how experience and location influence earnings in the data science field.

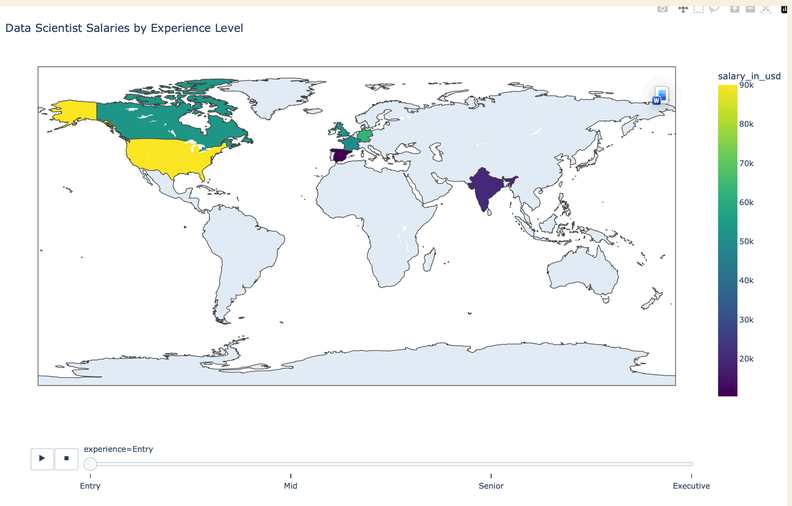
Introduction

The demand for data scientists has grown globally, but salaries vary depending on location and experience. Understanding these differences is important for students and professionals making career decisions. This project uses real-world salary data to analyze how earnings change across countries and experience levels. By exploring these patterns, the goal is to better understand how location and experience influence salaries in the data science field.

Visualizations







Interactive Map:Click here

Research Questions

- Which countries offer the highest salaries for data scientists?
- How does salary vary across experience levels (Entry, Mid, Senior, Executive)?
- How does company size influence data scientist salaries?
- How do experience level and company size together affect salary?

Results

The United States and Canada have the highest salaries, especially at higher experience levels.

· Salaries increase significantly from Entry to Executive levels, making experience the strongest factor.

· Larger companies tend to offer higher salaries, especially at senior levels, though there is overlap across company sizes.

· The impact of company size becomes more noticeable as experience increases.

Conclusion

Overall, the results show that data scientist salaries are influenced by multiple factors, including experience level, location, and company size. Salaries increase significantly with experience, while countries like the United States and Canada offer higher pay. Larger companies also tend to pay more, especially at higher experience levels. These findings show that salary outcomes are shaped by a combination of factors rather than a single variable.

Limitations

- Salaries are not adjusted for cost of living, which may affect comparisons between countries.
- Median salary was used to reduce the impact of extreme values.
- Countries with very small sample sizes were filtered out to provide more reliable comparisons.

References

1. Bhatia, R. (2022, June 15). Data Science Job Salaries. Kaggle. https://www.kaggle.com/datasets/ruchi798/data-science-job-salaries